

Recitation 5

Conditional Mean and Average Treatment Effect Review and Practice Problems

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October 9, 2026

Reading: Probability and Statistics for Data Science, Sections 7.8–7.9.

1 Conditional mean

Fix an observed value x , then average $\tilde{y}$ using its conditional distribution. Assuming the relevant means exist and are finite,

$$\mu_{\tilde{y}|\tilde{x}}(x) := \mathbb{E}[\tilde{y} \mid \tilde{x} = x] = \begin{cases} \sum y p_{\tilde{y}|\tilde{x}}(y \mid x), & \text{discrete } \tilde{y}, \\ \int_{-\infty}^{\infty} y f_{\tilde{y}|\tilde{x}}(y \mid x) dy, & \text{continuous } \tilde{y}. \end{cases}$$

When both variables are discrete, normalize by $p_{\tilde{x}}(x) > 0$: $p_{\tilde{y}|\tilde{x}}(y \mid x) = p_{\tilde{x},\tilde{y}}(x, y) / p_{\tilde{x}}(x)$.

A function versus a random variable. $\mu_{\tilde{y}|\tilde{x}}(x)$ is a number once x is fixed. The *conditional mean function* maps x to that number. Plugging in the random input gives the *conditional mean* $\mu_{\tilde{y}|\tilde{x}}(\tilde{x}) = \mathbb{E}[\tilde{y} \mid \tilde{x}]$, which is a random variable.

Iterated expectation (the tower law). Average the conditional means using the probabilities of the groups:

$$\mathbb{E}[\tilde{y}] = \mathbb{E}[\mu_{\tilde{y}|\tilde{x}}(\tilde{x})].$$

For discrete $\tilde{x}$ this becomes

$$\mathbb{E}[\tilde{y}] = \sum_x p_{\tilde{x}}(x) \mu_{\tilde{y}|\tilde{x}}(x).$$

For continuous $\tilde{x}$, replace the sum by an integral against $f_{\tilde{x}}(x)$. Group means get equal weight only when group probabilities are equal.

From data. For observed pairs $X = \{(x_i, y_i) : i = 1, \dots, n\}$, let $I_x = \{i : x_i = x\}$. When $|I_x| > 0$, estimate the conditional mean by the within-group average $|I_x|^{-1} \sum_{i \in I_x} y_i$. For a continuous predictor, exact repeats may be unavailable; one can average nearby observations or fit a model for the conditional mean.

2 Average treatment effect

Let $\tilde{t} \in \{0, 1\}$ denote treatment. Each unit has potential outcomes $\widetilde{\text{po}}_0$ under control and $\widetilde{\text{po}}_1$ under treatment. The observed outcome satisfies **consistency**:

$$\tilde{y} = (1 - \tilde{t})\widetilde{\text{po}}_0 + \tilde{t}\widetilde{\text{po}}_1.$$

Only the potential outcome for the received treatment is observed. The population **average treatment effect** is

$$\text{ATE} = \mathbb{E}[\widetilde{\text{po}}_1] - \mathbb{E}[\widetilde{\text{po}}_0] = \mathbb{E}[\widetilde{\text{po}}_1 - \widetilde{\text{po}}_0].$$

Observed contrast and randomization. The difference visible in the observed groups is

$$\Delta_{\text{obs}} := \mu_{\tilde{y}|\tilde{t}}(1) - \mu_{\tilde{y}|\tilde{t}}(0).$$

In general this compares different populations. If $\widetilde{\text{po}}_r \perp\!\!\!\perp \tilde{t}$ for $r = 0, 1$, and both treatment groups have positive probability, then

$$\mu_{\tilde{y}|\tilde{t}}(r) = \mathbb{E}[\widetilde{\text{po}}_r \mid \tilde{t} = r] = \mathbb{E}[\widetilde{\text{po}}_r], \quad \Delta_{\text{obs}} = \text{ATE}.$$

Random assignment provides this independence. Differences of *sample* means still have sampling error; they estimate the population contrast.

Adjustment for a discrete confounder. Let $\tilde{c}$ be a pre-treatment characteristic, such as smoking status. Write $\mu_{\tilde{y}|\tilde{c},\tilde{t}}(c, r) = \mathbb{E}[\tilde{y} \mid \tilde{c} = c, \tilde{t} = r]$. Assume:

1. Consistency as above.
2. Conditional independence: $\widetilde{\text{po}}_r \perp\!\!\!\perp \tilde{t} \mid \tilde{c}$ for $r = 0, 1$.
3. Positivity: $0 < \mathbb{P}(\tilde{t} = 1 \mid \tilde{c} = c) < 1$ for every c with $p_{\tilde{c}}(c) > 0$.

Iterated expectation gives the adjustment formula:

$$\text{ATE} = \sum_c p_{\tilde{c}}(c) \left[\mu_{\tilde{y}|\tilde{c},\tilde{t}}(c, 1) - \mu_{\tilde{y}|\tilde{c},\tilde{t}}(c, 0) \right].$$

Indeed, for either r ,

$$\mathbb{E}[\widetilde{\text{po}}_r] = \sum_c p_{\tilde{c}}(c) \mathbb{E}[\widetilde{\text{po}}_r \mid \tilde{c} = c] = \sum_c p_{\tilde{c}}(c) \mu_{\tilde{y}|\tilde{c},\tilde{t}}(c, r).$$

The second equality uses conditional independence and consistency.

Choose the weights before calculating. Adjustment uses the *same target-population weights* $p_{\tilde{c}}(c)$ for both treatments. The raw group mean instead uses $p_{\tilde{c}|\tilde{t}}(c \mid r)$, which may differ across treatment groups. For a target consisting of all study participants, estimate the common weights using the pooled counts.

Practice problems

Four core problems; Problem 5 is optional. The indicated times are for independent work before discussion.

Problem 1: Life expectancy (Exercise 7.11)

[5 minutes]

In the model in Exercise 7.11, age at death has mean 37.5 years. First-year deaths have probability 0.25 and are recorded as age 0. Let $\tilde{a}$ denote the recorded age at death, and let $\tilde{s}$ equal 1 if a person survives the first year and 0 otherwise.

- (a) Write $\mathbb{E}[\tilde{a}]$ as a weighted average of $\mu_{\tilde{a}|\tilde{s}}(0)$ and $\mu_{\tilde{a}|\tilde{s}}(1)$.
- (b) Find the mean age at death among people who survive the first year. Explain why this differs from the mean for the whole population.

Problem 2: A conditional mean from a model**[8 minutes]**

Adapted from the uniform–Bernoulli exercise in the Fall 2025 archive, mine/r4.

Let $\tilde{u} \sim \text{Uniform}(0, 1)$, and conditional on $\tilde{u} = u$, let $\tilde{x} \sim \text{Bernoulli}(u)$. Here $\tilde{u}$ is explicitly random.

- (a) Find $\mathbb{P}(\tilde{x} = 1)$ and the conditional densities $f_{\tilde{u}|\tilde{x}}(u \mid 1)$ and $f_{\tilde{u}|\tilde{x}}(u \mid 0)$.
- (b) Compute the conditional mean function $g(x) := \mu_{\tilde{u}|\tilde{x}}(x)$ for $x \in \{0, 1\}$.
- (c) Give the possible values and probabilities of the random variable $g(\tilde{x})$. Verify that $\mathbb{E}[g(\tilde{x})] = \mathbb{E}[\tilde{u}]$.

Problem 3: Coffee and life expectancy**[10 minutes]**

Exercise 7.17. The observational study gives these counts and average lifespans:

	Number of people		Mean lifespan (years)	
	Coffee	No coffee	Coffee	No coffee
Smokers	80	20	68	70
Non-smokers	40	60	82	80

Let $\tilde{t} = 1$ mean drinking coffee, $\tilde{c} = 1$ mean smoking, and $\tilde{y}$ denote lifespan. Use all study participants as the target population.

- (a) Compute the unadjusted difference in sample means, coffee minus no coffee (the exercise's "observed ATE").
- (b) Estimate the ATE after adjusting for smoking. Assume $\widetilde{\text{po}}_r \perp\!\!\!\perp \tilde{t} \mid \tilde{c}$ for $r = 0, 1$, as well as consistency and positivity. Show the weights you use.
- (c) Explain the change after adjustment. Compare the two within-group differences with the adjusted estimate.

Problem 4: Rolling until the first six**[8 minutes]**

Roll a fair six-sided die independently until the first 6 appears. Let $\tilde{n}$ be the *total* number of rolls, including the final 6. Let A be the event that at least one 1 appears before that first 6.

- (a) Find $\mathbb{E}[\tilde{n}]$.
- (b) Find $\mathbb{E}[\tilde{n} \mid A]$, the expected total number of rolls given that at least one 1 appears before stopping.
- (c) Find $\mathbb{E}[\tilde{n} \mid A^c]$, the expected total number of rolls given that no 1 appears before stopping.

Problem 5 (backup): Splitting a Poisson total

Optional if time allows. Suggested working time: 5–7 minutes.

Let $\tilde{x}$ and $\tilde{y}$ be independent $\text{Poisson}(\lambda)$ random variables, with $\lambda > 0$, and define $\tilde{s} = \tilde{x} + \tilde{y}$.

- (a) For any integer $n \geq 0$, find $\mu_{\tilde{x}|\tilde{s}}(n) = \mathbb{E}[\tilde{x} \mid \tilde{s} = n]$. *Hint:* use symmetry and $\tilde{x} + \tilde{y} = n$ under the condition.
- (b) Write the random conditional mean $\mu_{\tilde{x}|\tilde{s}}(\tilde{s})$ and use iterated expectation to recover $\mathbb{E}[\tilde{x}]$.

Source: adapted from Problem 3(b) of wyt/Recitation 4.pdf in the Fall 2025 archive.